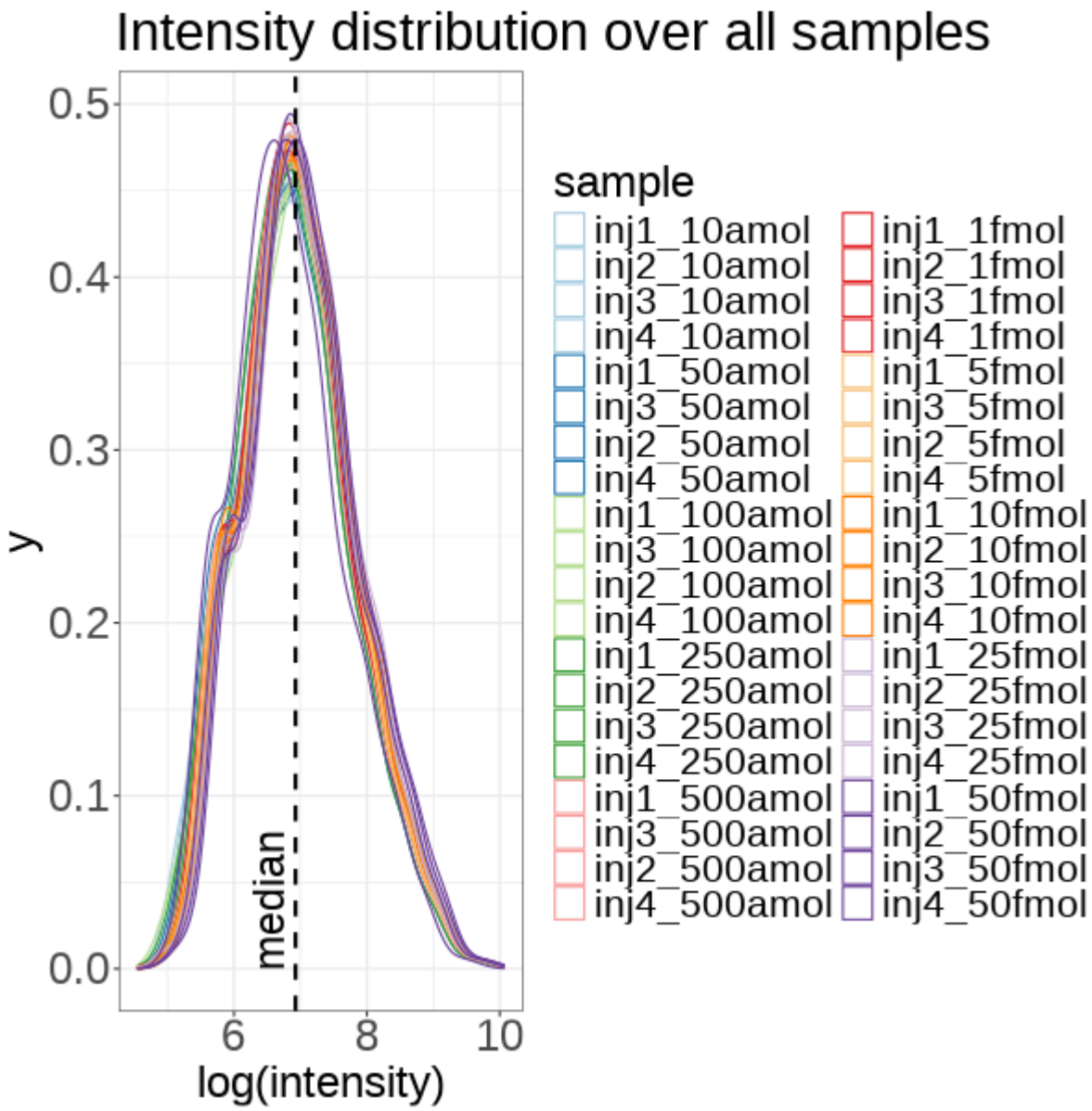
Quality control of Parsed_proteins_set.txt

Analysis

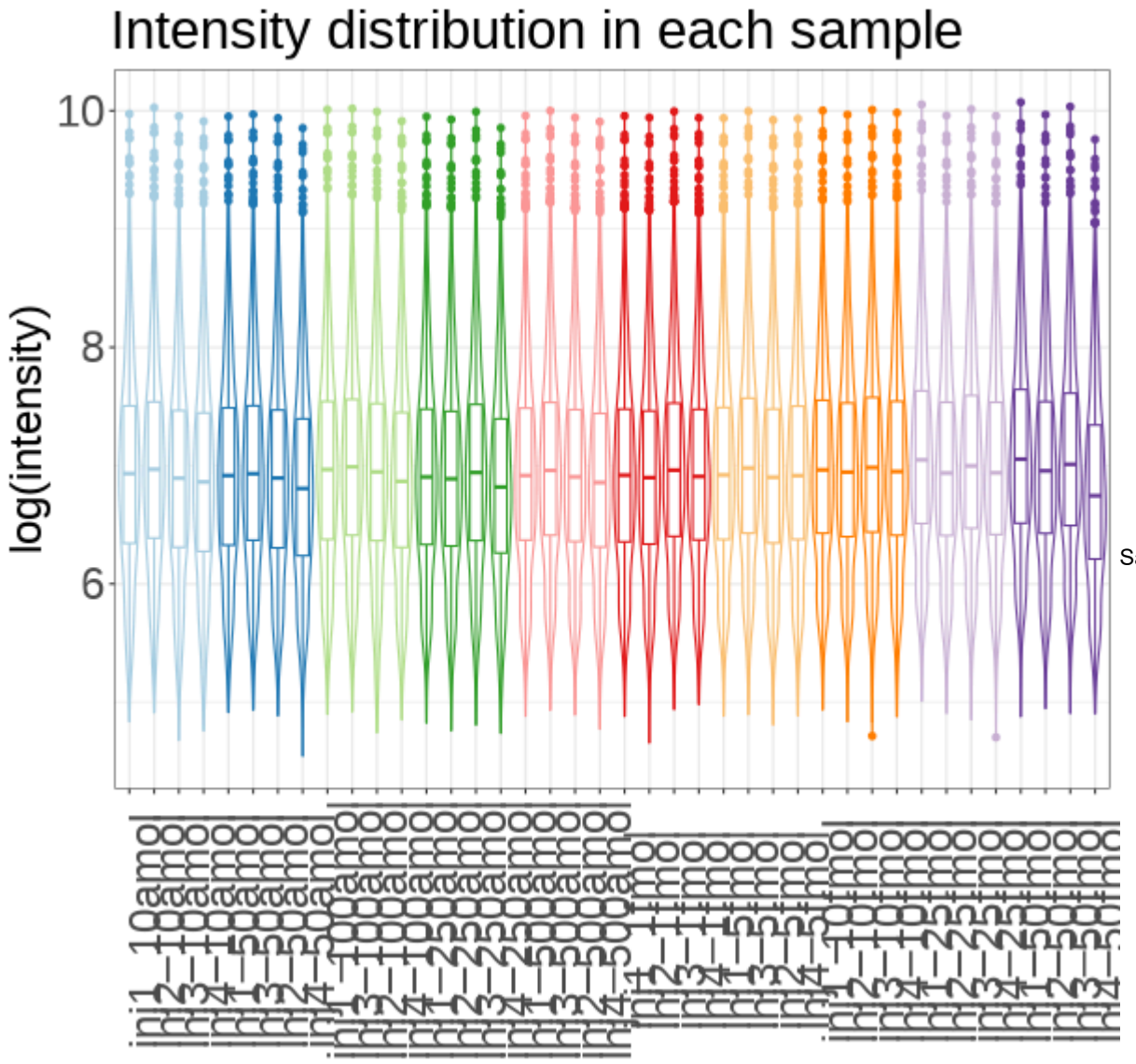
R-Analysis Version	2.11
File	D:_portable\4.2.0-Portable_with_QP-proteomics-DA_proteins_set.txt
Origin File	
Date	26-11-13 11:05:26

Raw data visualization

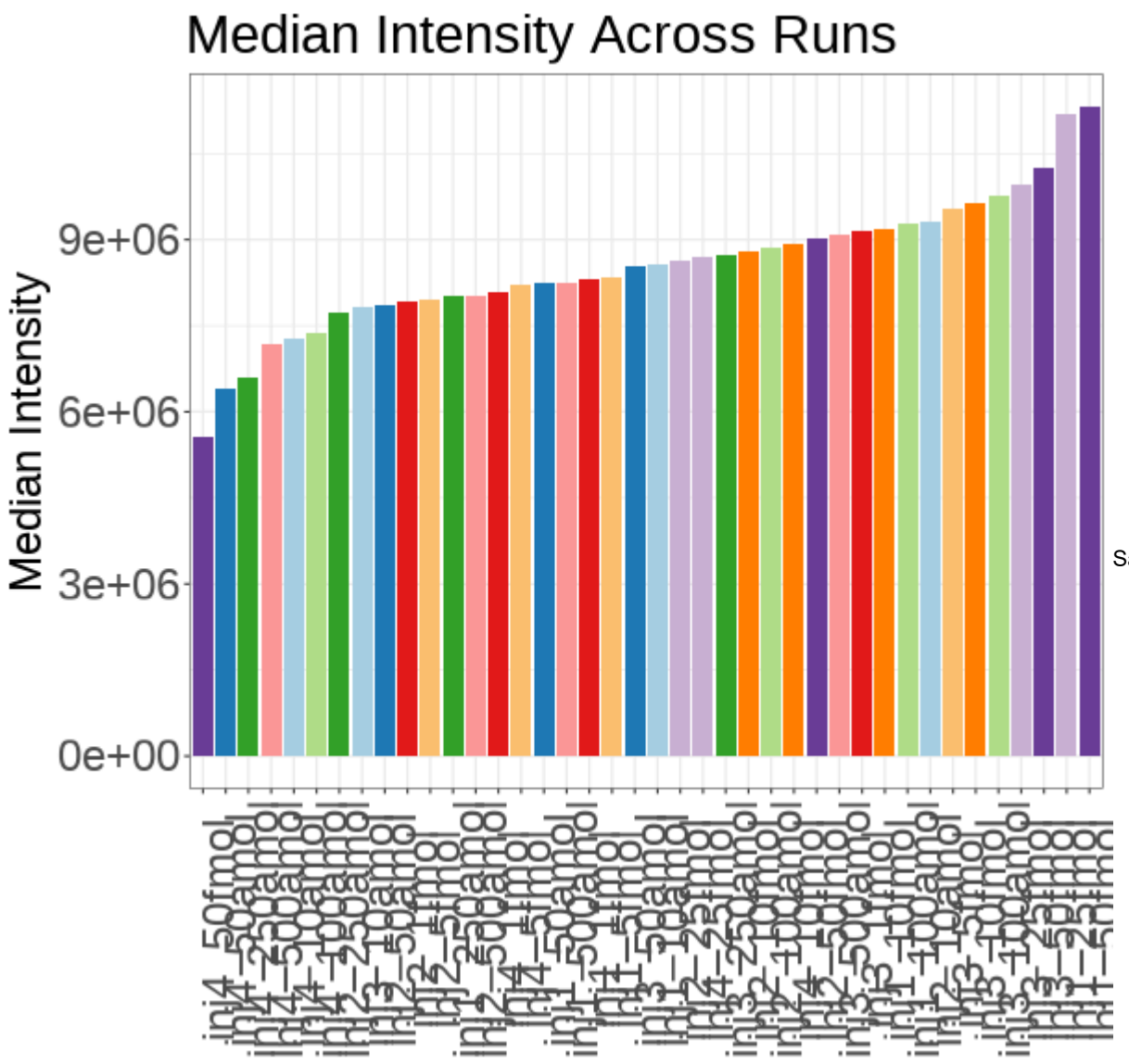


Save figure

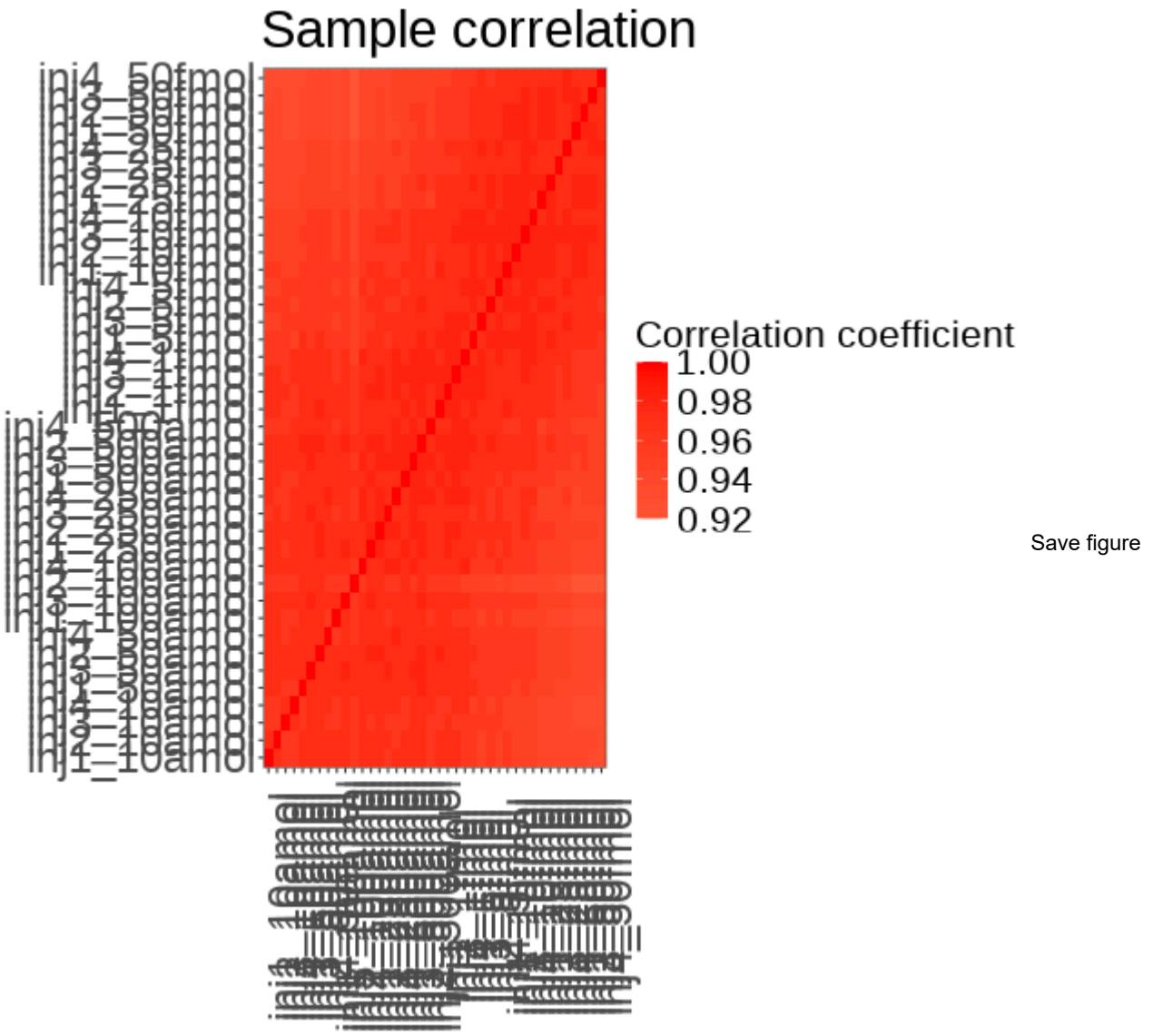
Plots density over the log intensity of each sample

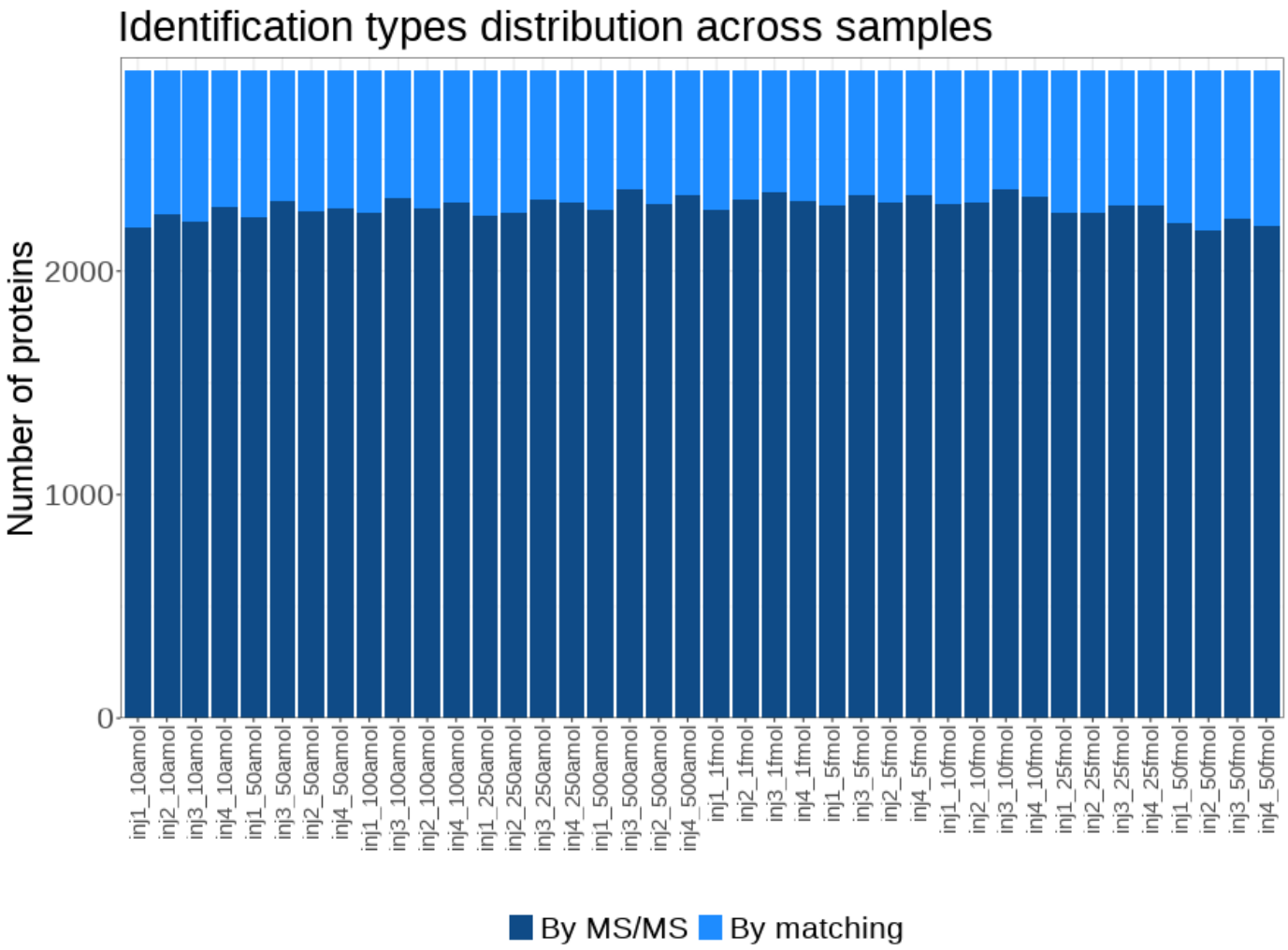


Save figure


Median Intensity Across Runs
Median Intensity
9e+06
6e+06
3e+06
0e+00

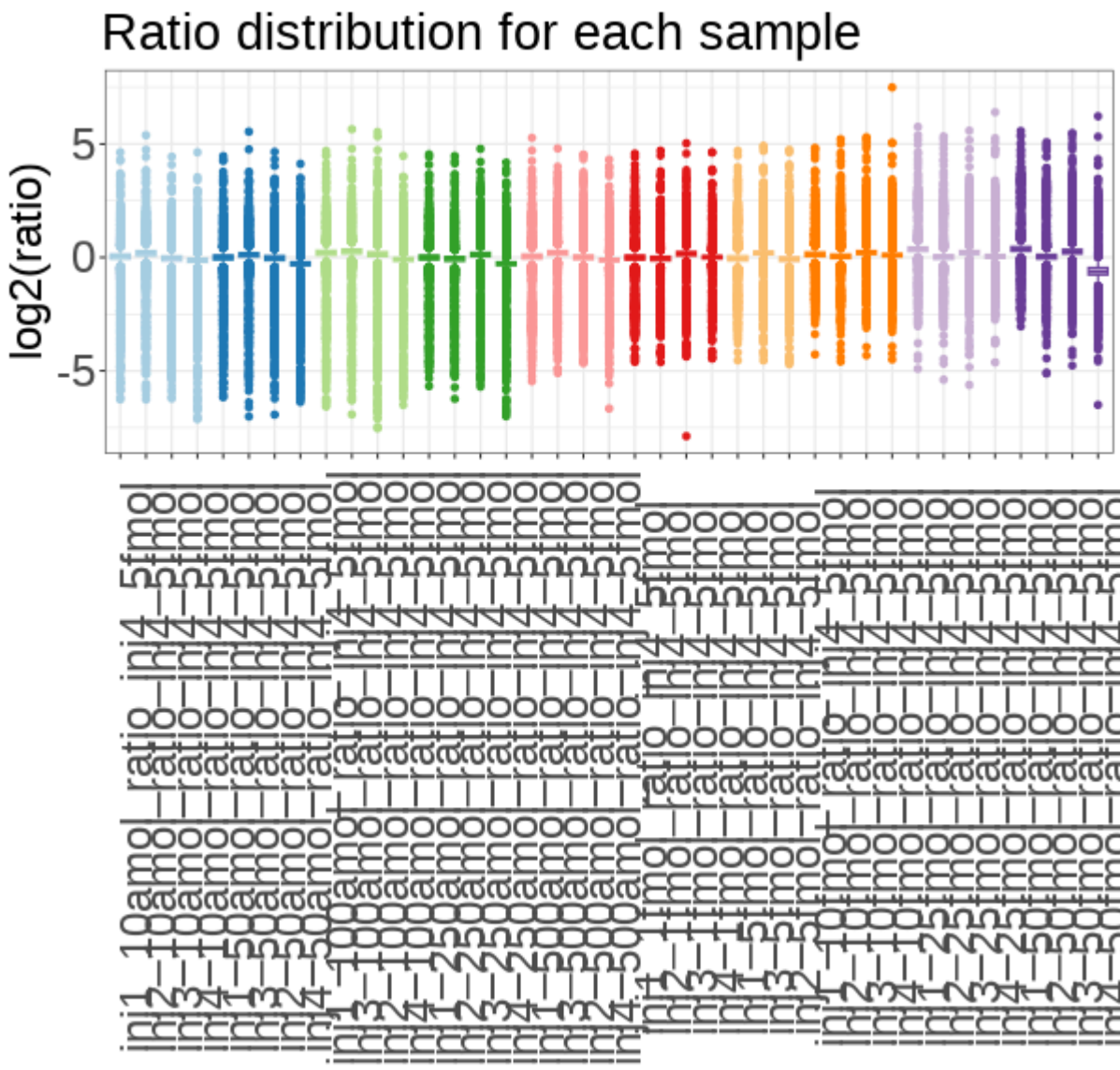
Save figure


Sample correlation
Correlation coefficient
1.00
0.98
0.96
0.94
0.92
Save figure

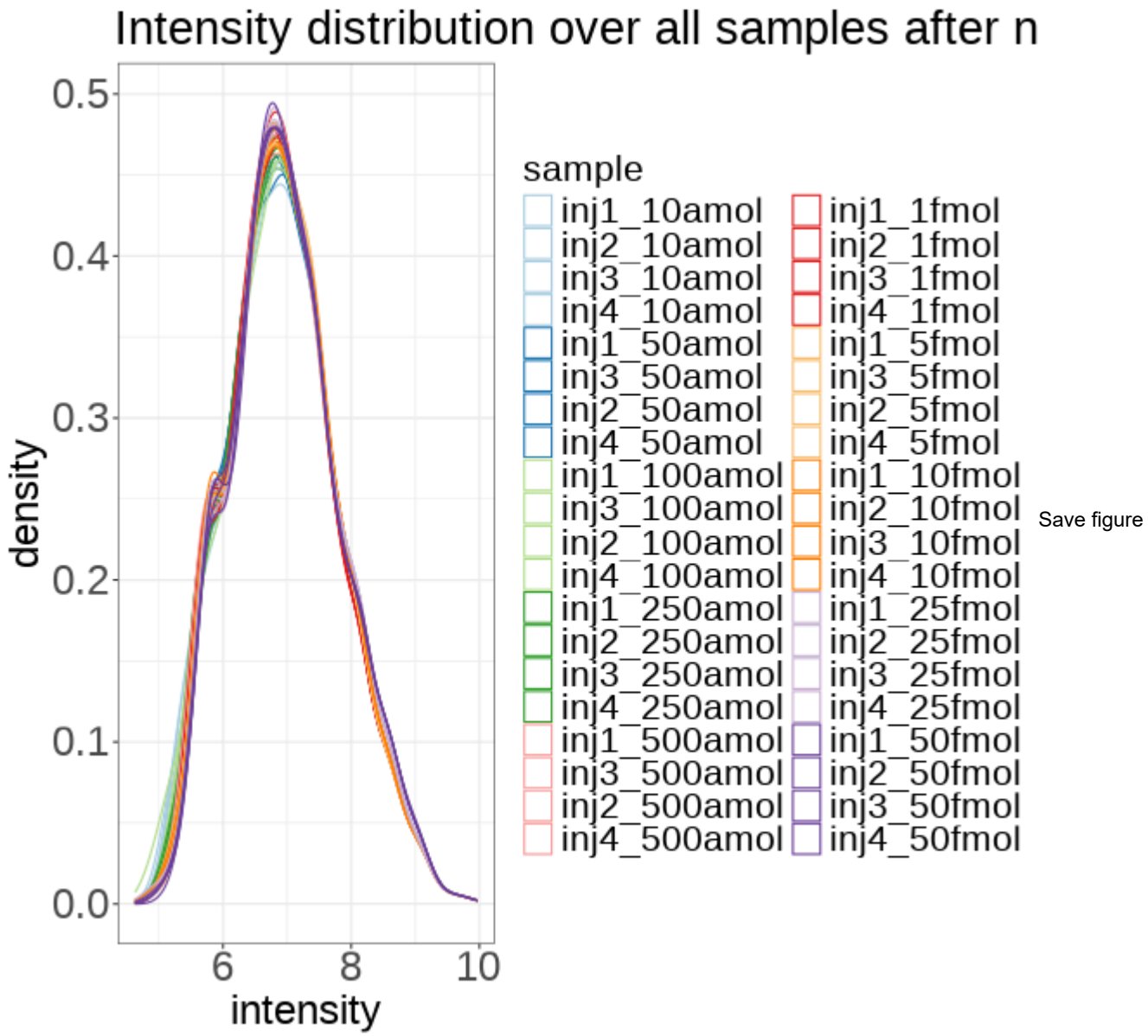


Save figure

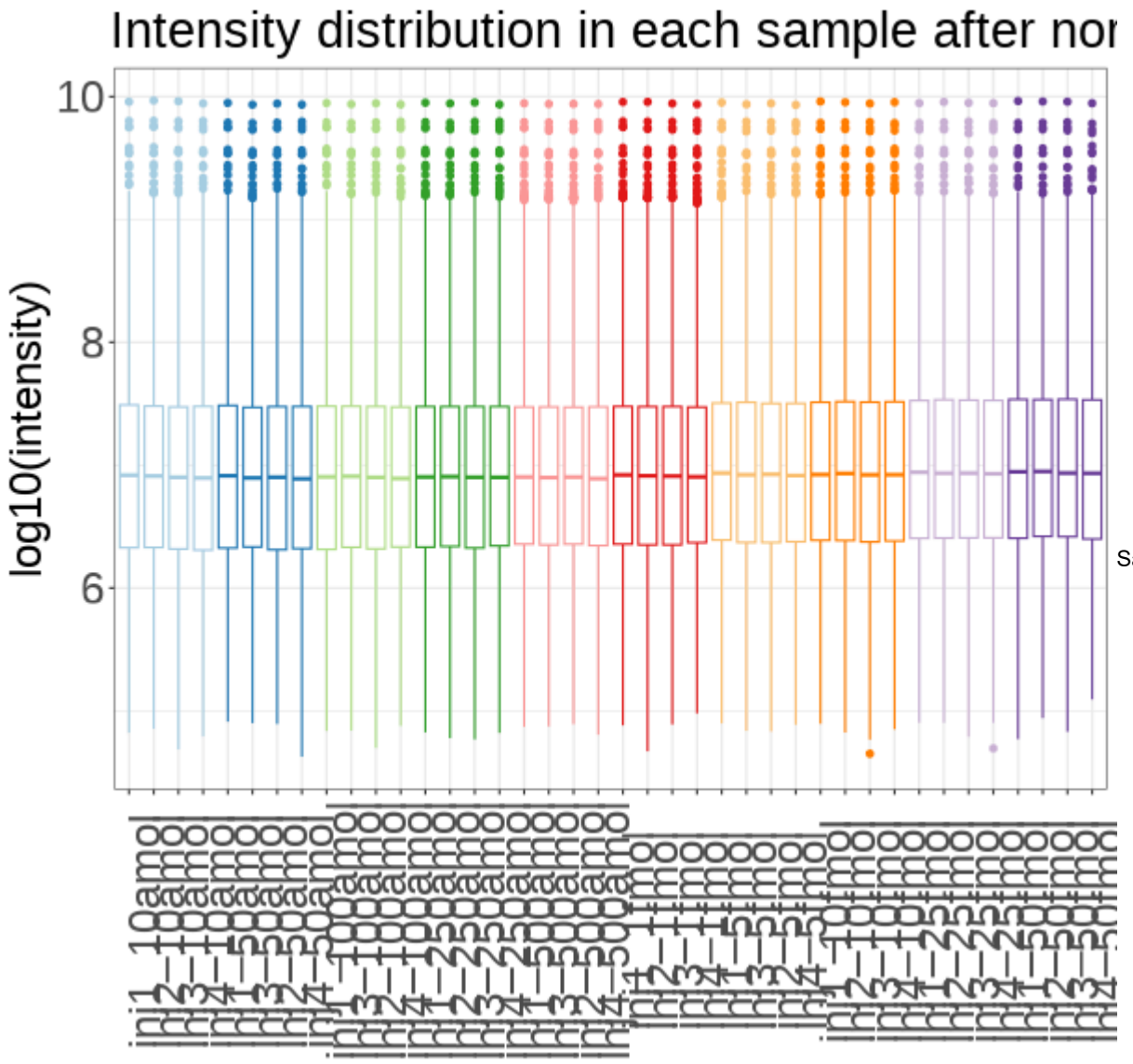
Normalization



Save figure


Intensity distribution over all samples after n
density
0.5
0.4
0.3
0.2
0.1
0.0
6
8
10
intensity
sample
inj1_10amol
inj2_10amol
inj3_10amol
inj4_10amol
inj1_50amol
inj3_50amol
inj2_50amol
inj4_50amol
inj1_100amol
inj3_100amol
inj2_100amol
inj4_100amol
inj1_250amol
inj2_250amol
inj3_250amol
inj4_250amol
inj1_500amol
inj3_500amol
inj2_500amol
inj4_500amol
inj1_1fmol
inj2_1fmol
inj3_1fmol
inj4_1fmol
inj1_5fmol
inj3_5fmol
inj2_5fmol
inj4_5fmol
inj1_10fmol
inj2_10fmol
inj3_10fmol
inj4_10fmol
inj1_25fmol
inj2_25fmol
inj3_25fmol
inj4_25fmol
inj1_50fmol
inj2_50fmol
inj3_50fmol
inj4_50fmol

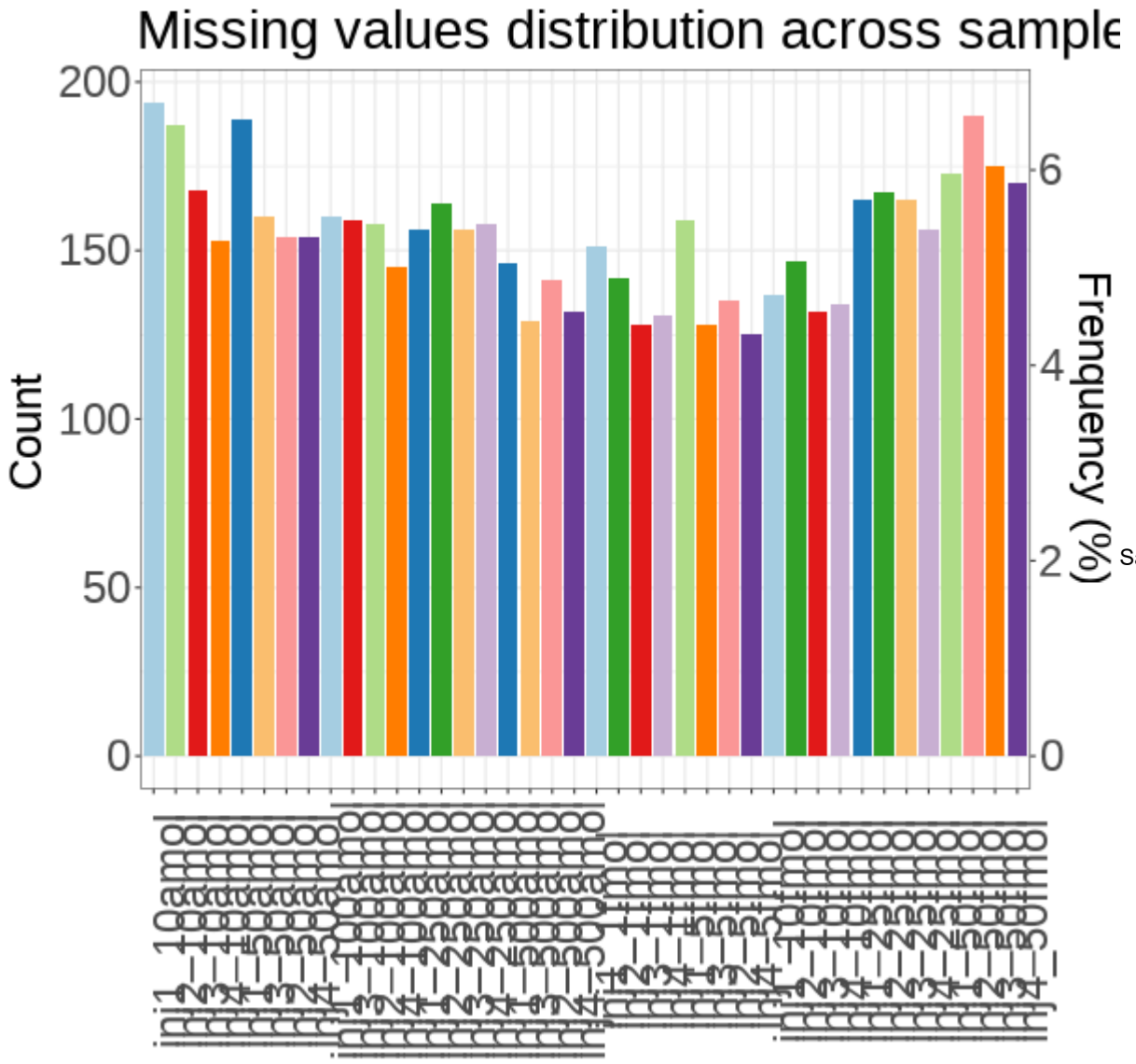
Save figure



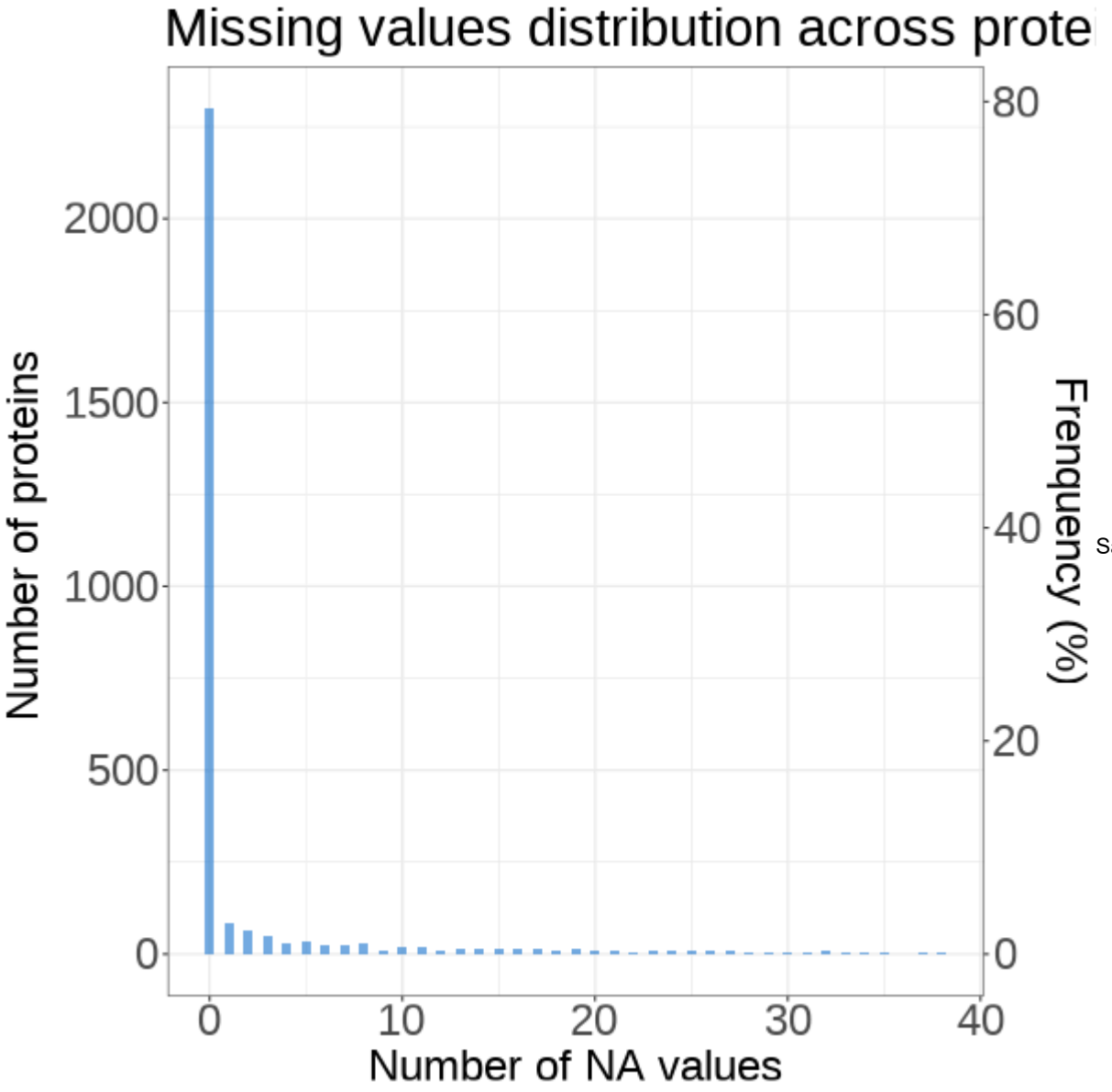
Save figure

Normalization method : each intensity is expressed relatively to a reference intensity, as a ratio of intensities. Intensities of the sample with the lowest number of missing values are used as reference intensities. Intensities are then centered by substracting the median of the ratios for each sample.

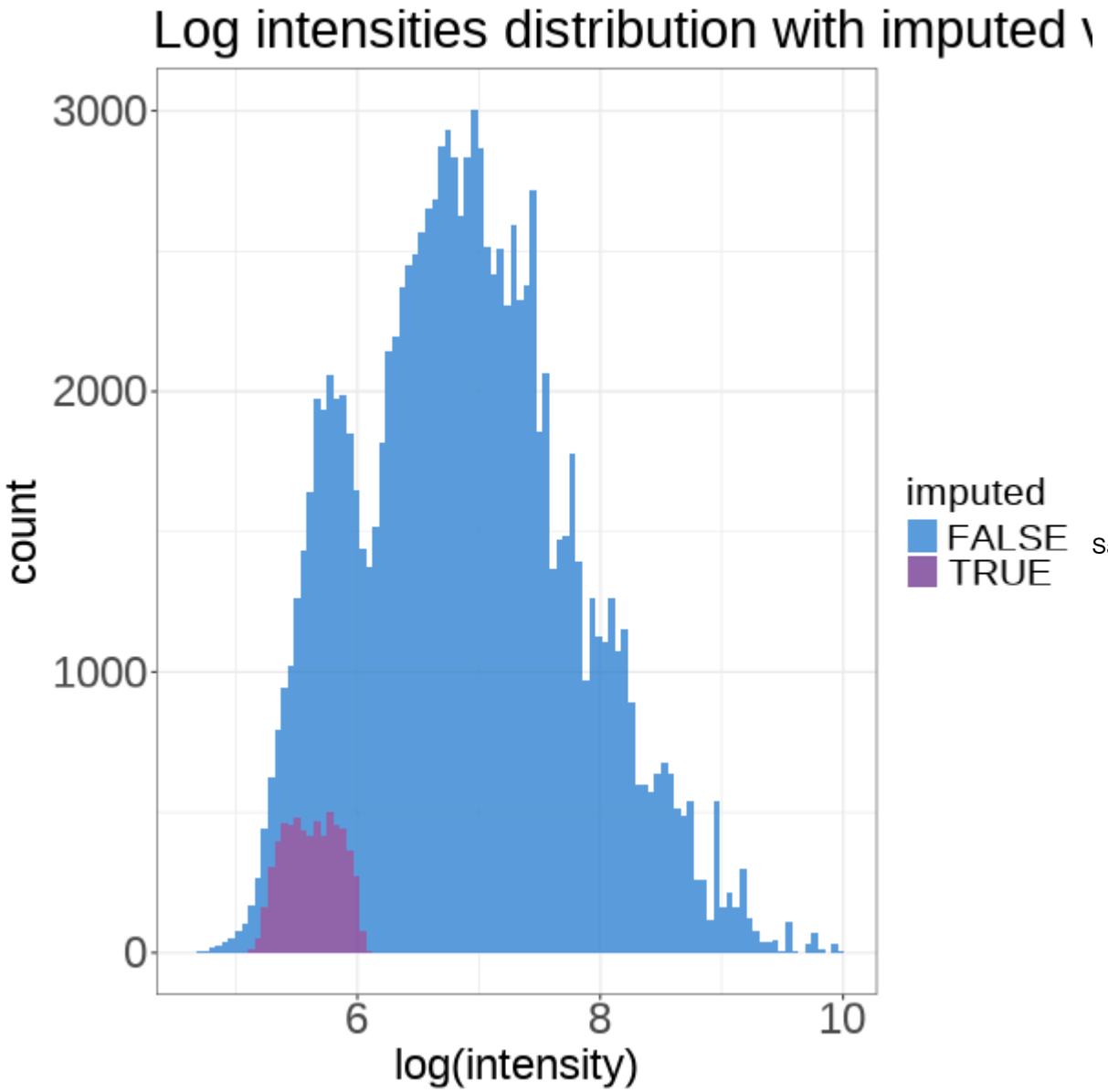
Missing values imputation


Missing values distribution across sample
Count
Frenquency (%)
200
150
100
50
0
6
4
2
0

Save figure

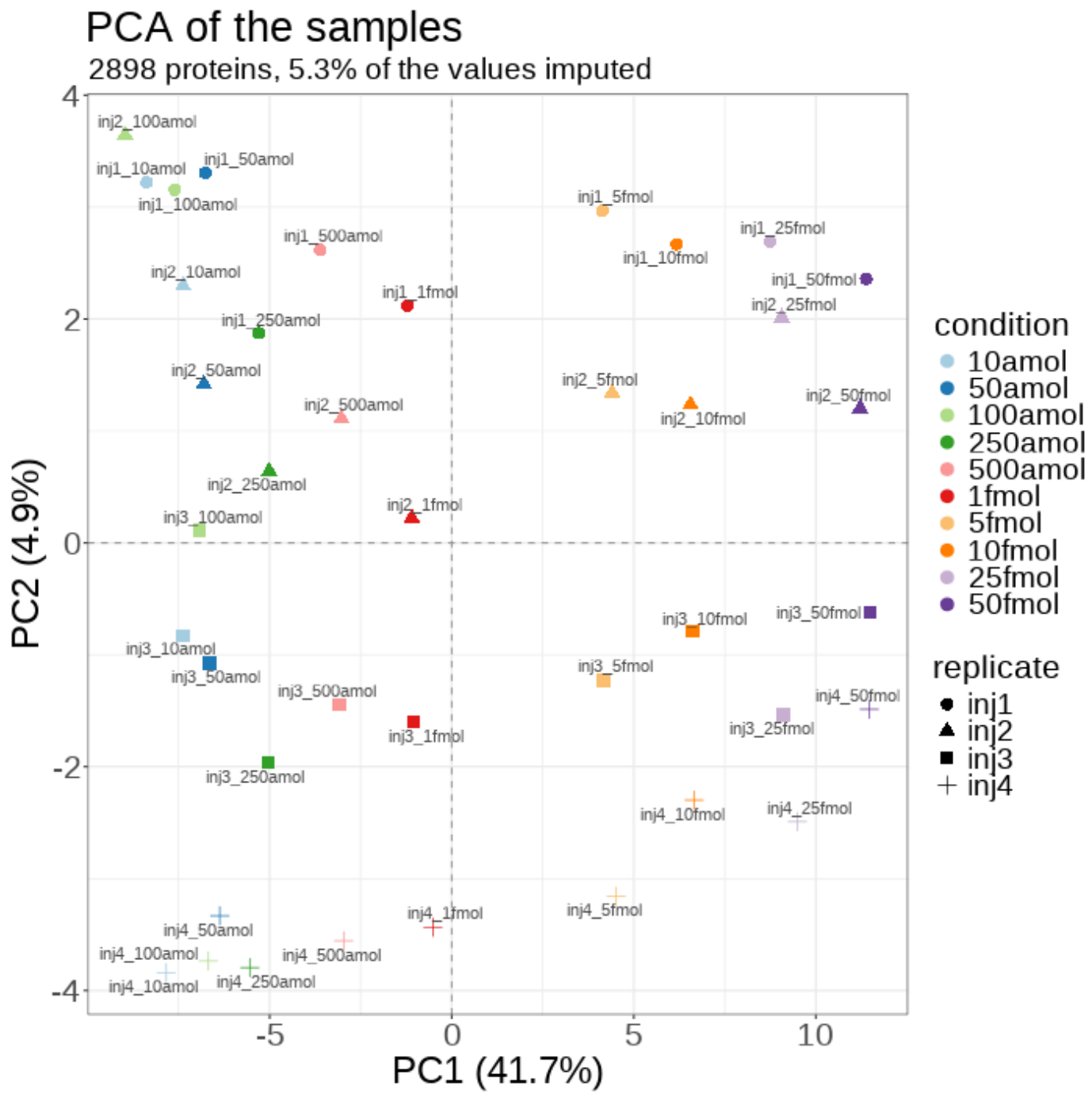

Missing values distribution across prote
Number of proteins
2000
1500
1000
500
0
80
60
40
20
0
Frenquency (%)
0
10
20
30
40
Number of NA values

Save figure



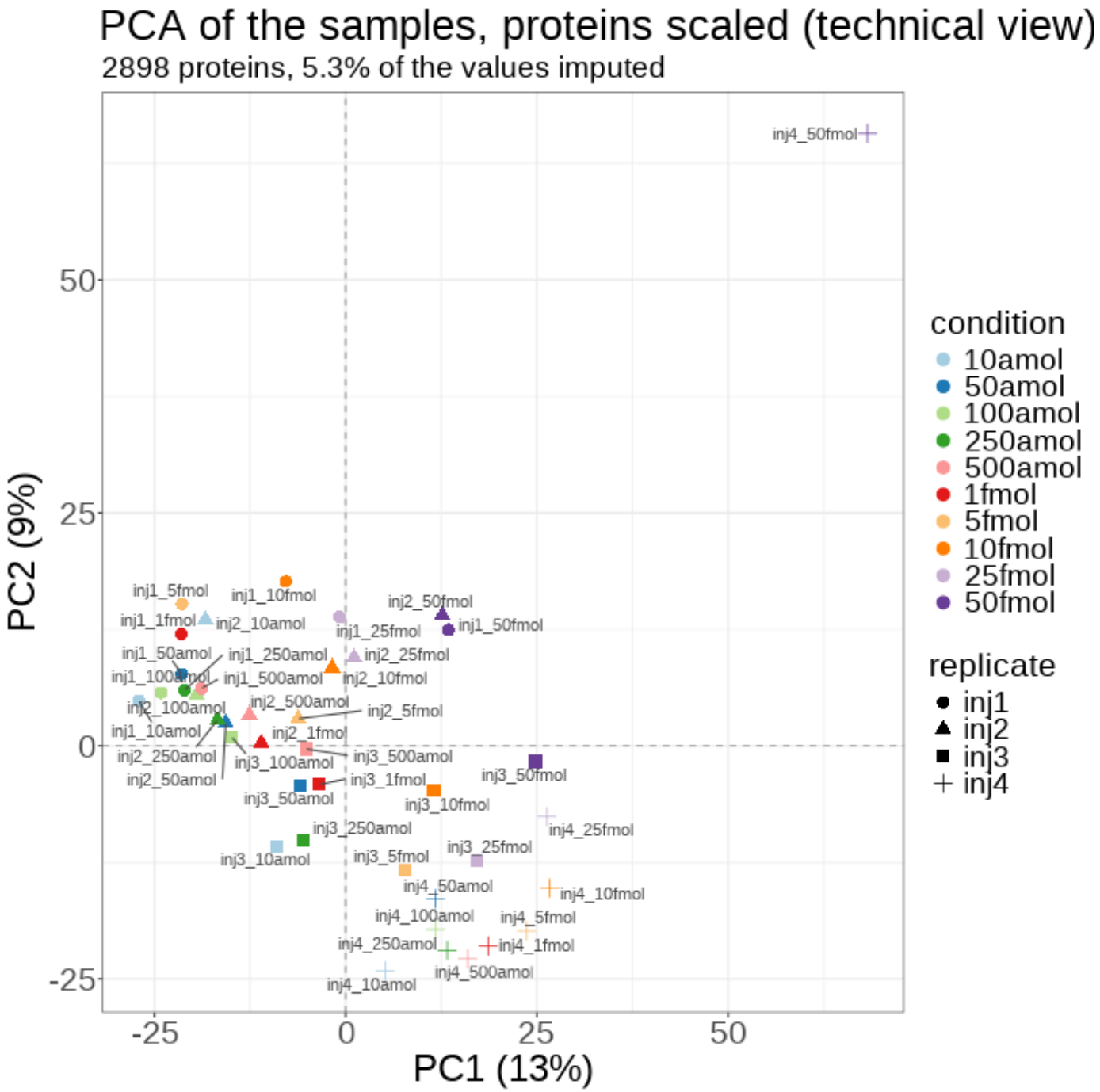
Missing values are imputed with the noise model: values are drawn around the noise level of each sample (the 5% percentile of its observed intensities) with a standard deviation of 0.81 log2, estimated on 6701 low-intensity protein measurements within conditions. The imputation uses a fixed seed and is reproducible.

Dimension reduction



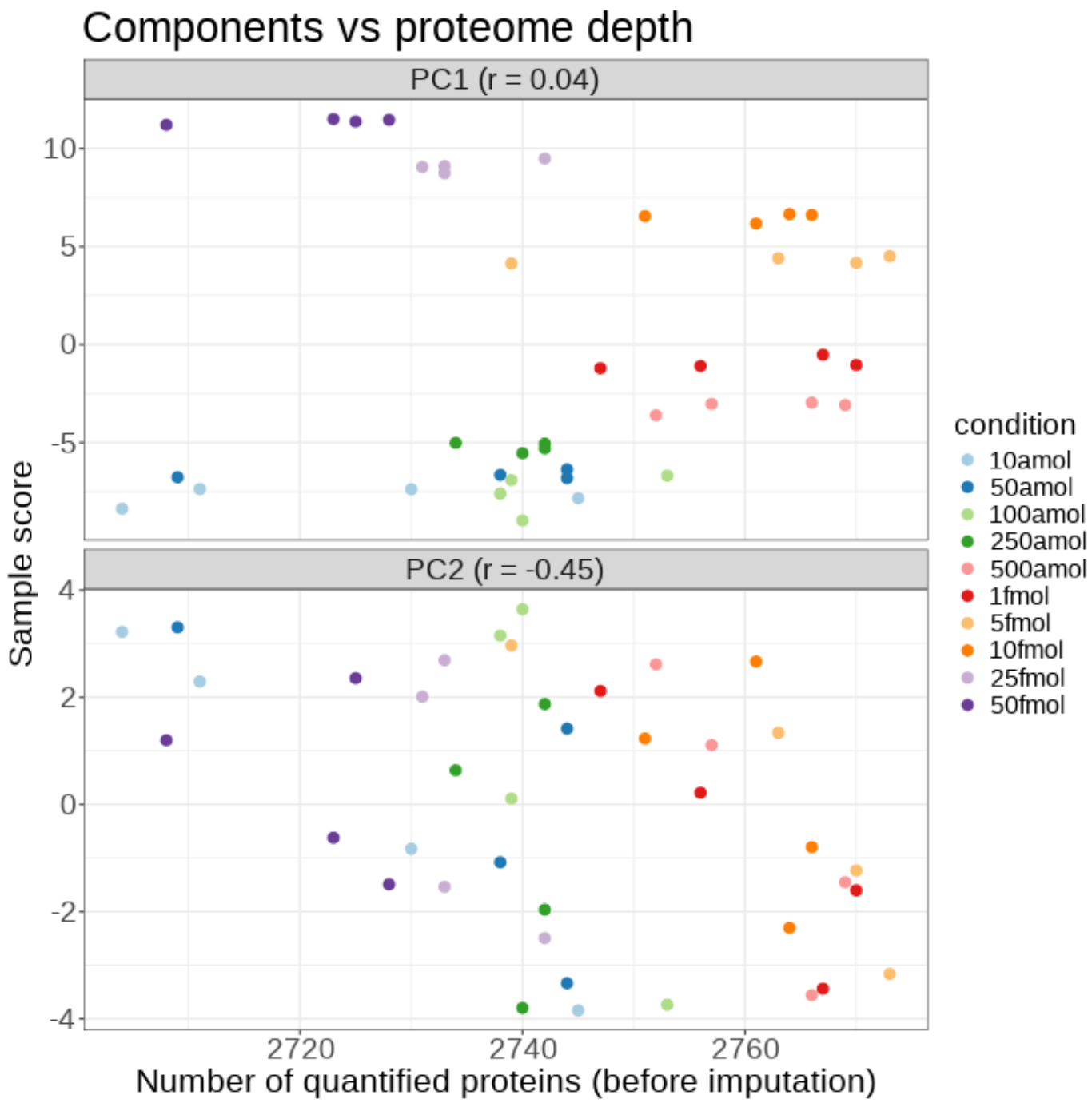
Save figure

Log10 intensities, each protein centered. Proteins vary in weight according to their variance, so the components reflect the largest changes between samples.



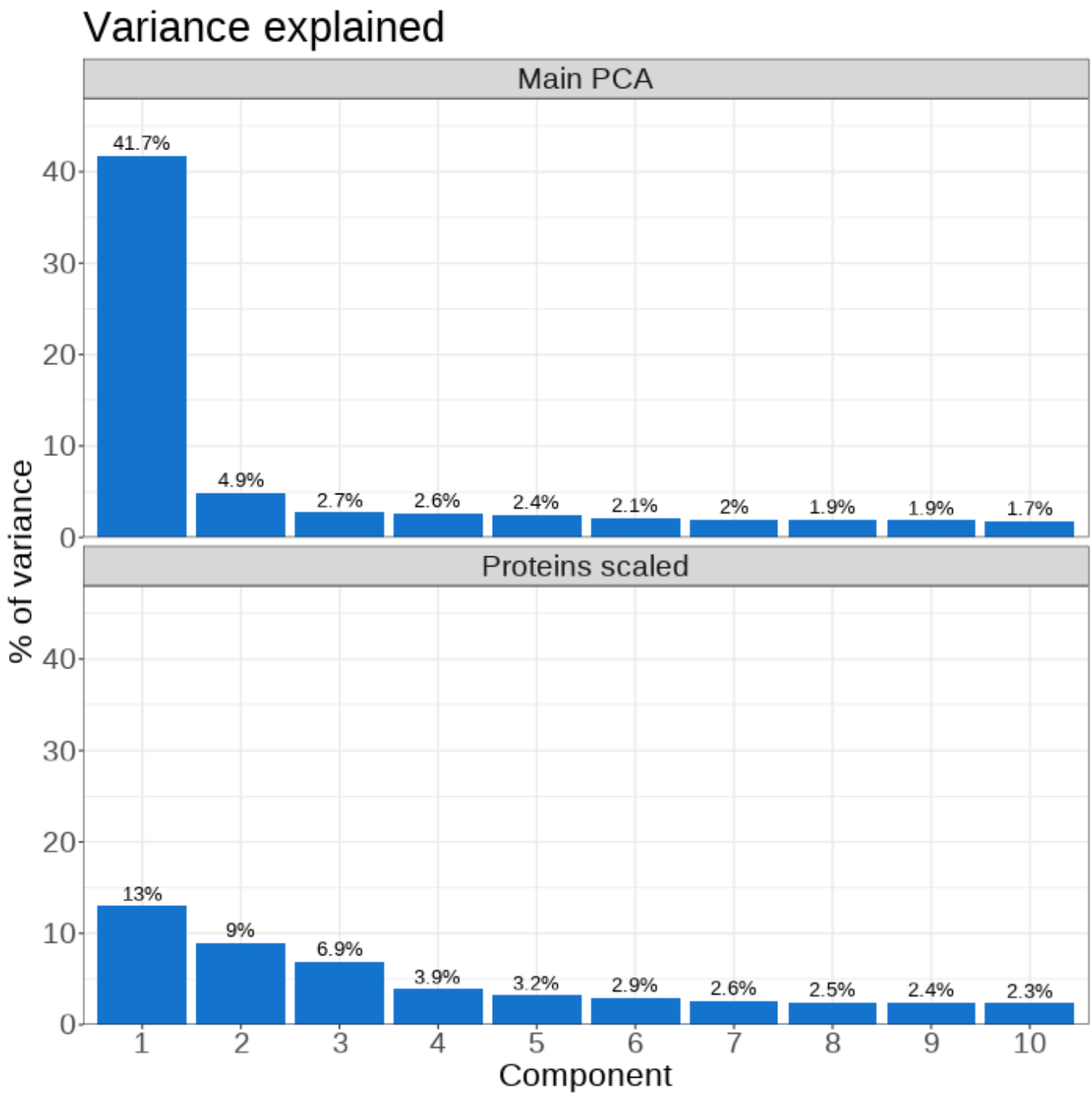
Save figure

Each protein is also scaled to unit variance, so all proteins weigh the same, including the many proteins that do not change. Their technical noise then has as much weight as the biological changes: this view helps to detect technical effects (injection order, batches), not biological ones.



Save figure

Sample scores of the main PCA against the number of proteins quantified in each sample. A strong correlation (|r| close to 1) means that the component mostly reflects differences in proteome depth (and therefore imputed values) rather than biology.



Save figure

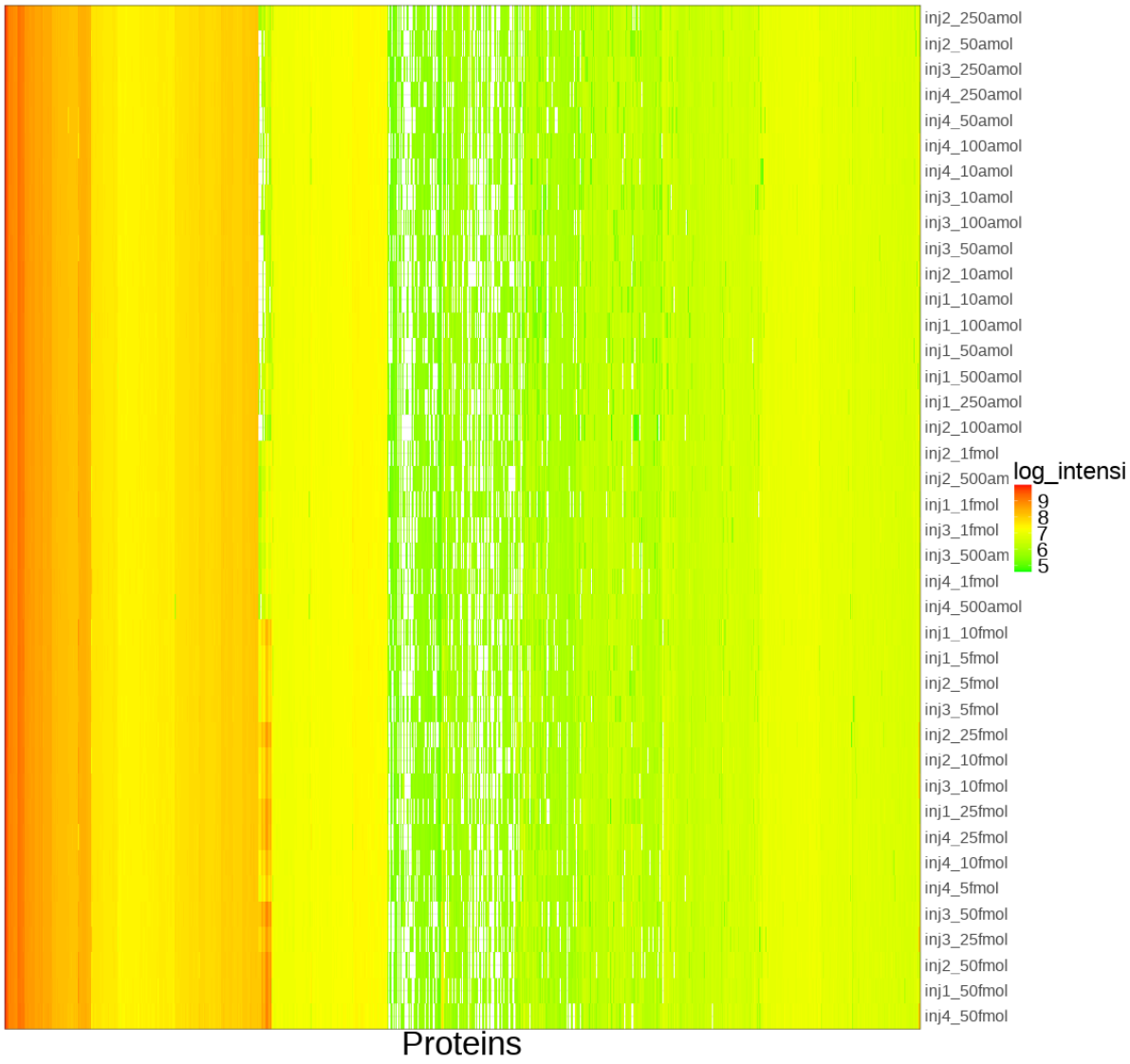
Proteins with the largest loadings on PC1 and PC2 of the main PCA (the sign gives the direction along the component)

Component	Rank	Protein	Loading

Component	Rank	Protein	Loading
PC1	1	NQO2_HUMAN_UPS	0.161
	2	UBE2C_HUMAN_UPS	0.159
	3	CYB5_HUMAN_UPS	0.158
	4	MYG_HUMAN_UPS	0.154
	5	NQO1_HUMAN_UPS	0.154
	6	CAH1_HUMAN_UPS	0.153
	7	GSTA1_HUMAN_UPS	0.153
	8	IGF2_HUMAN_UPS	0.152
	9	CAH2_HUMAN_UPS	0.148
	10	SYHC_HUMAN_UPS	0.148
PC2	1	RT07_YEAST	-0.152
	2	SC61G_YEAST	0.149
	3	HIBCH_YEAST	0.148
	4	RPF2_YEAST	0.147
	5	SA155_YEAST	-0.143
	6	AVT1_YEAST	-0.120
	7	SEC15_YEAST	-0.112
	8	ADT1_YEAST	0.108
	9	FKH1_YEAST	-0.101
	10	PSK1_YEAST	0.100

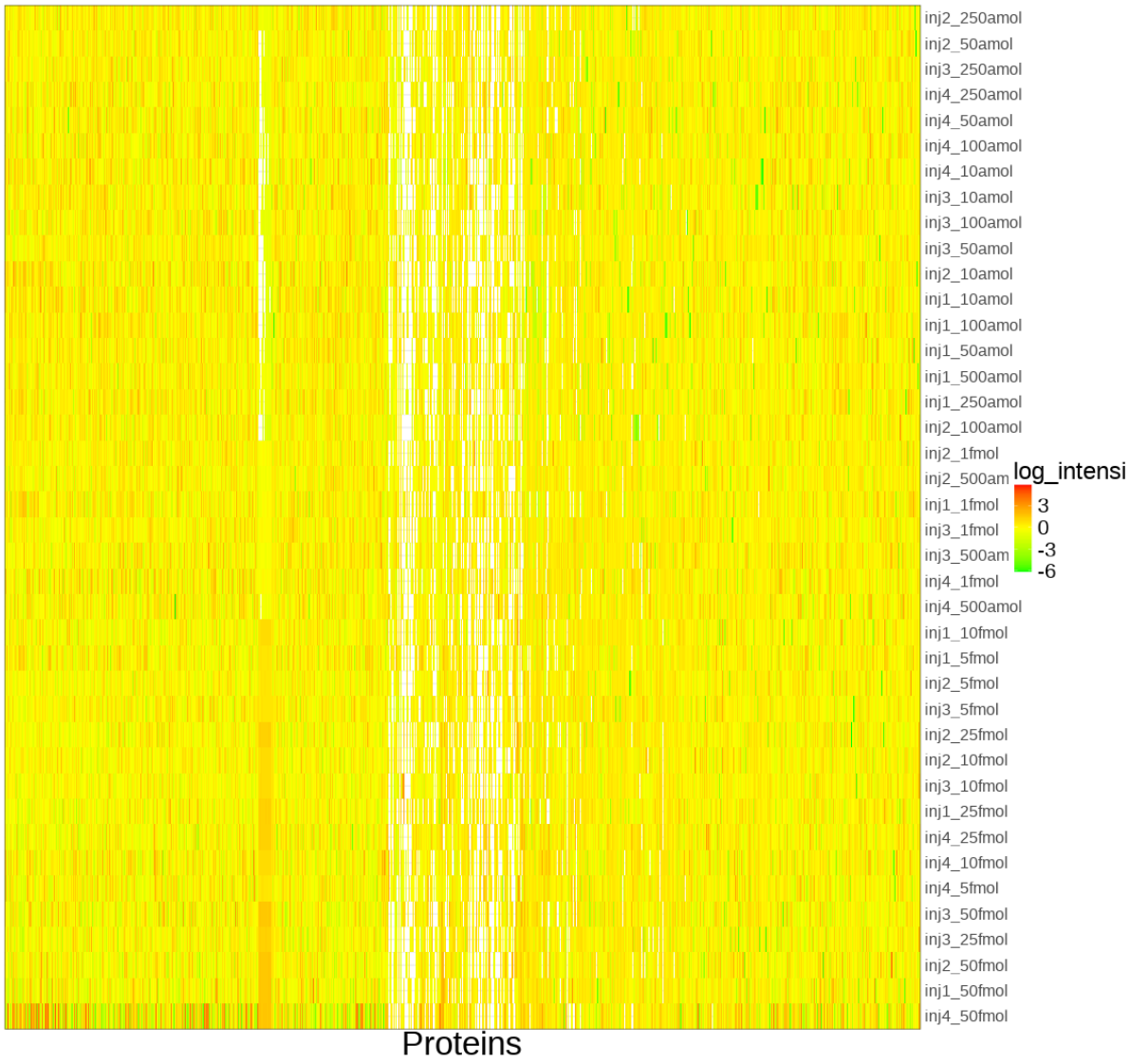
Hierarchical clustering

Log Intensities Heatmap



Save figure

Z-Scored Log Intensities Heatmap



Save figure